Computational PDEs: Classical Approaches and Physics-Informed Learning

Day 3 — Doubly Connected Domains

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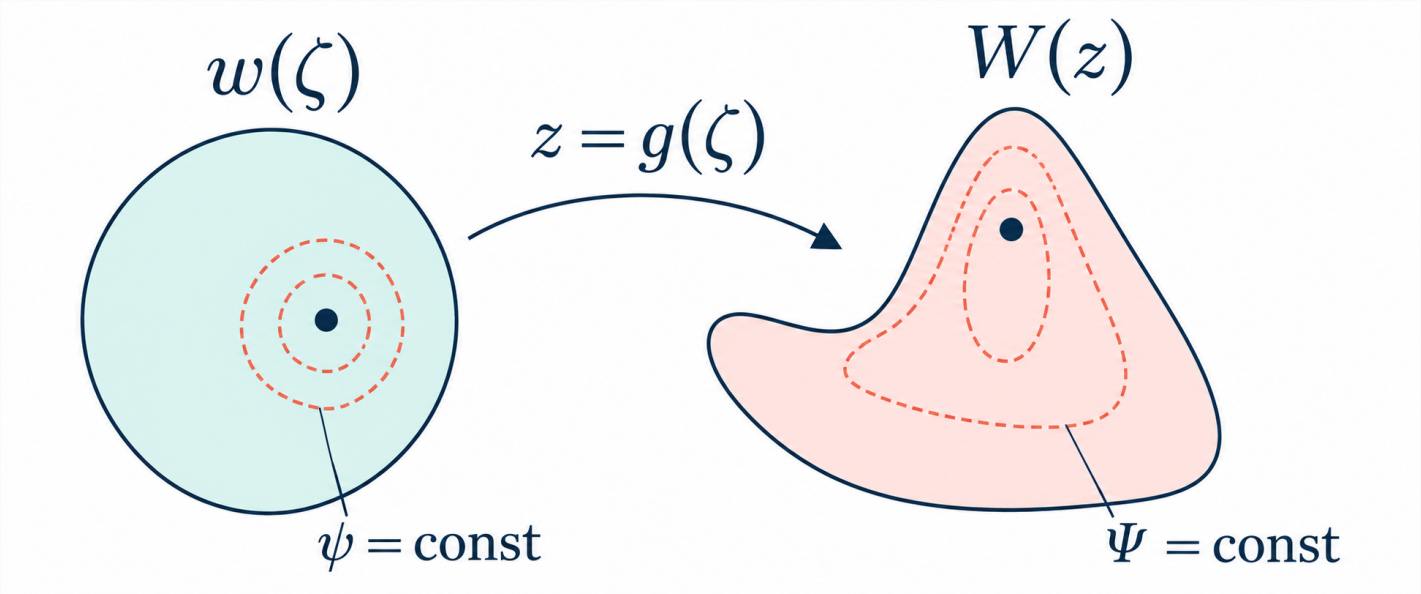
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Outline

1. Input: a list of circles. One hole is an annulus
2. The key idea is "**Schottky–Klein prime functions**"
3. The product $P(\zeta, \rho)$, truncated at $N_{\max}$
4. The unit test: the inversion identity at sample points
5. The network learns the regular part $\hat{w}$. The product is the check

From Tuesday

Tuesday placed one reflected point. A hole is not removed by a map to the disk.



Today

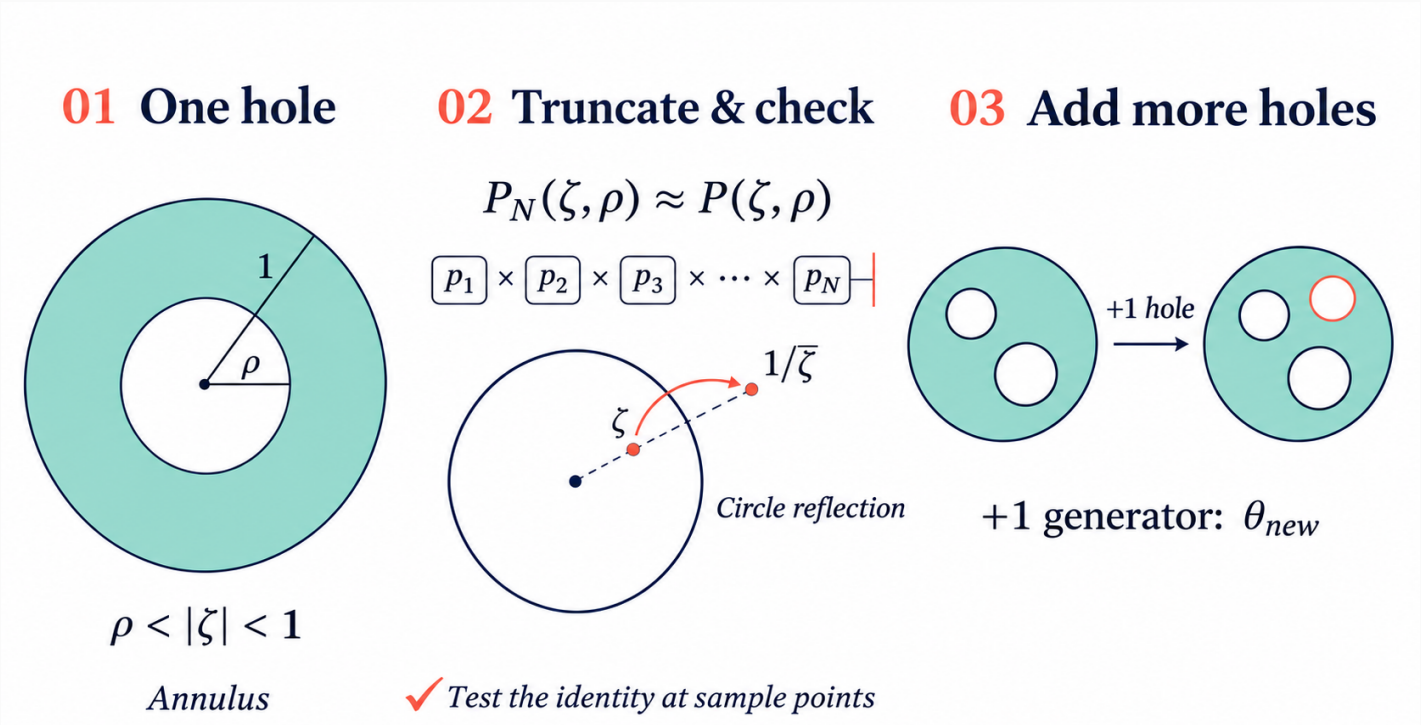
The input is a list of circles. Reflections that do not stop are a product. The network learns only the regular part of the vortex.

We are going to extend this theory to multiply connected domains.

Aims

After today you should be able to

1. Annulus, with inner radius ρ, as the domain with one hole;
2. truncate a special function $P(\zeta, \rho)$ and evaluate the inversion identity at sample points;
3. say that further holes append one generator (Advanced topic)



A $\log$ singularity inside the unit disk

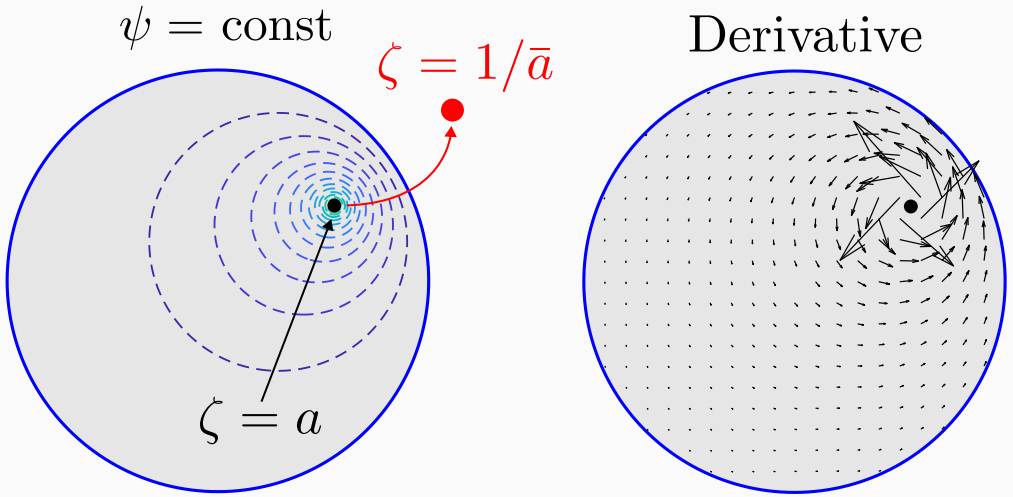
Suppose a $\log$-type singularity sits at $\zeta = a$ inside the unit disk. Impose

$$\psi = 0 \text{ on } \partial\mathbb{D} \tag{1}$$

($\mathbb{D}$: unit disk; one chosen level of ψ on the boundary). Then

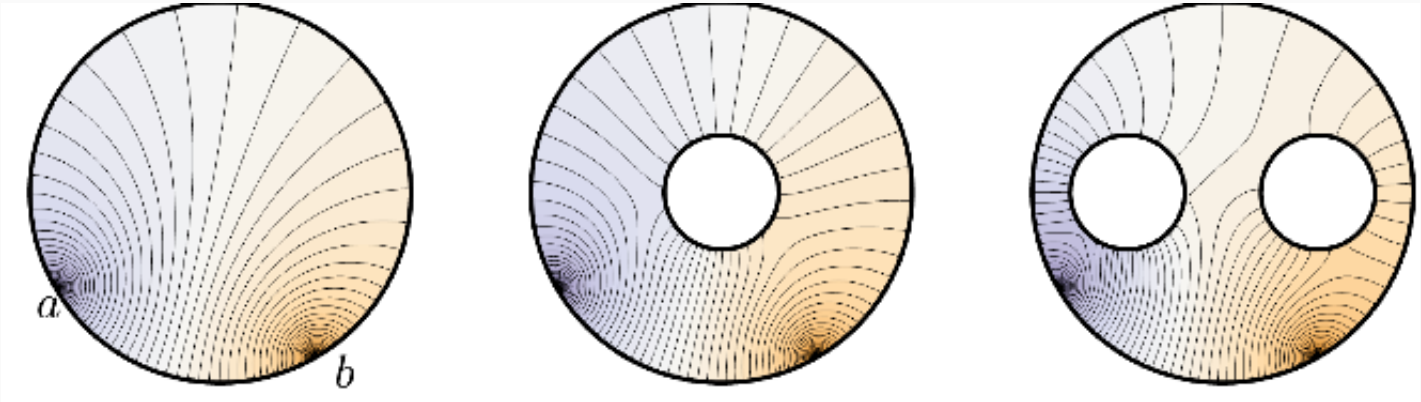
$$w(\zeta) = \phi + \mathrm{i}\psi = \frac{1}{2\pi\mathrm{i}} \log\left(\frac{\zeta - a}{|a|(\zeta - 1/\overline{a})}\right)$$

The **image point in the unit circle** is $\zeta = 1/\overline{a}$.



Riemann mapping theorem (multiply connected domains)

- The **Riemann mapping theorem** extends to **multiply connected domains**.
- There is a conformal map between a simply connected domain and the unit disk.
- A doubly connected domain corresponds to an **annulus** (a domain with one hole).
- A triply connected domain corresponds to a **unit disk with two holes**.



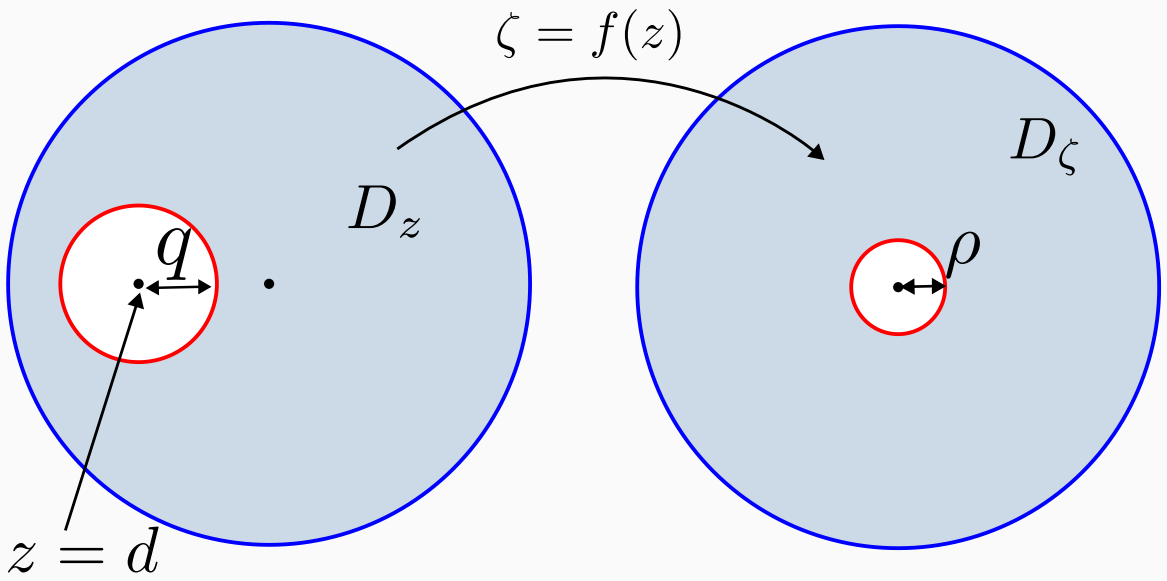
Goal from here

Build the theory on the domain above (a circular canonical domain).

From simply connected to doubly connected domains

Doubly connected domains

- For simply connected domains the unit disk was the starting point.
- For doubly connected domains it is the **annulus**.
- By an **automorphism**, the centre of the inner circle can be mapped to the origin. (Next page)
- The parameter of the annulus is the inner radius ρ (the modulus).
- Domain $D_\zeta = \{\zeta \, | \rho < |\zeta| < 1\}, \quad 0 < \rho < 1.$



Automorphisms

$$\zeta = e^{\mathrm{i}\alpha} \frac{z - z_0}{\overline{z_0} z - 1}, \quad \alpha \in \mathbb{R},\ d \in (D_z \backslash \partial D_z) \tag{2}$$

where

$$z_0 = \frac{1 + c^2 - r^2 - \sqrt{(1 + c^2 - r^2)^2 - 4c^2}}{2c} \tag{3}$$

Then a circle inside the unit disk, with center $z = c > 0$ and radius r, is mapped to a circle centered at the origin.

Automorphisms

Set $\alpha = 0$ and $z_0 \in \mathbb{R}$. Then $\zeta(z) = (z - z_0)/(z_0 z - 1)$ preserves the real axis. The circle $|z - c| = r$ lies in the unit disk and meets that axis at $a = c - r$ and $b = c + r$. The image is symmetric about the real axis, hence centered at the origin once $\zeta(a) = -\zeta(b)$:

$$\frac{a - z_0}{z_0 a - 1} = -\frac{b - z_0}{z_0 b - 1}. \tag{4}$$

Clearing the denominators gives $(a + b)z_0^2 - 2(1 + ab)z_0 + (a + b) = 0$. With $a + b = 2c$ and $ab = c^2 - r^2$, divide by $2c$:

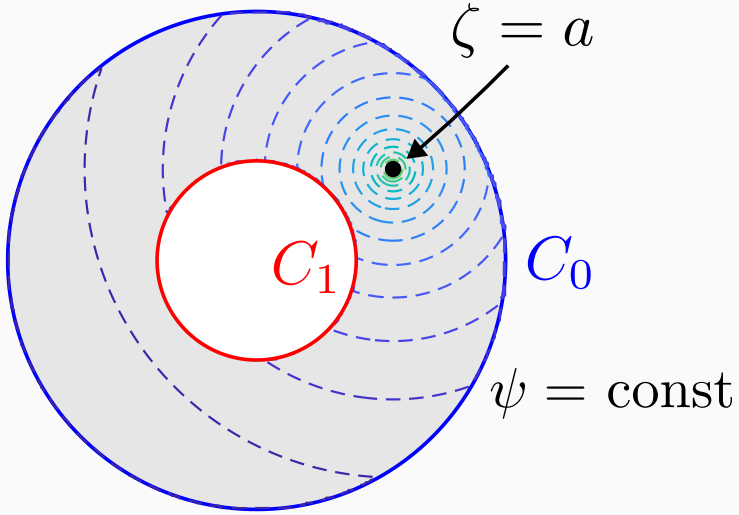
$$z_0^2 - \frac{1 + c^2 - r^2}{c} z_0 + 1 = 0. \tag{5}$$

The roots are reciprocal. The root in the unit disk is

$$z_0 = \frac{1 + c^2 - r^2 - \sqrt{(1 + c^2 - r^2)^2 - 4c^2}}{2c}. \tag{6}$$

Construction of complex potential

- Suppose a point vortex is located at $\zeta = a$ in the annulus.
- The unit circle C_0 and the inner circle C_1 are to be streamlines.
  $\rightarrow$ **There is no flux into the domain.**
- We want ψ to be constant on both C_0 and C_1.
  $\rightarrow$ Apply the idea of the method of images infinitely often.

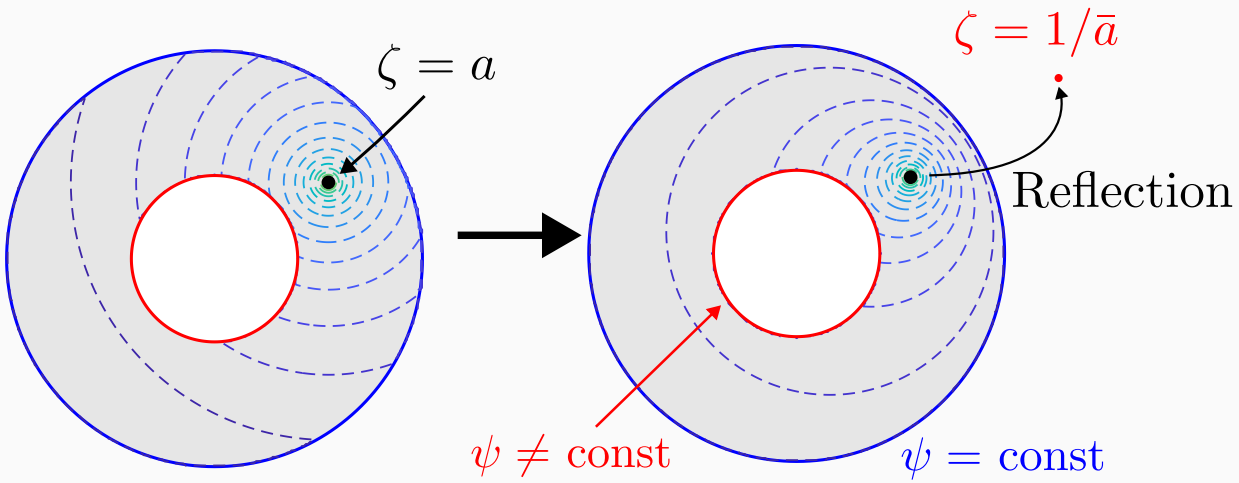


Construction of the complex potential

- For $\zeta = a$, consider the potential that includes the image $1/\overline{a}$ in the unit circle:

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}} (\log(\zeta - a) - \log(\zeta - 1/\overline{a})) \tag{7}$$

- The boundary condition is satisfied on C_0, but not on C_1.

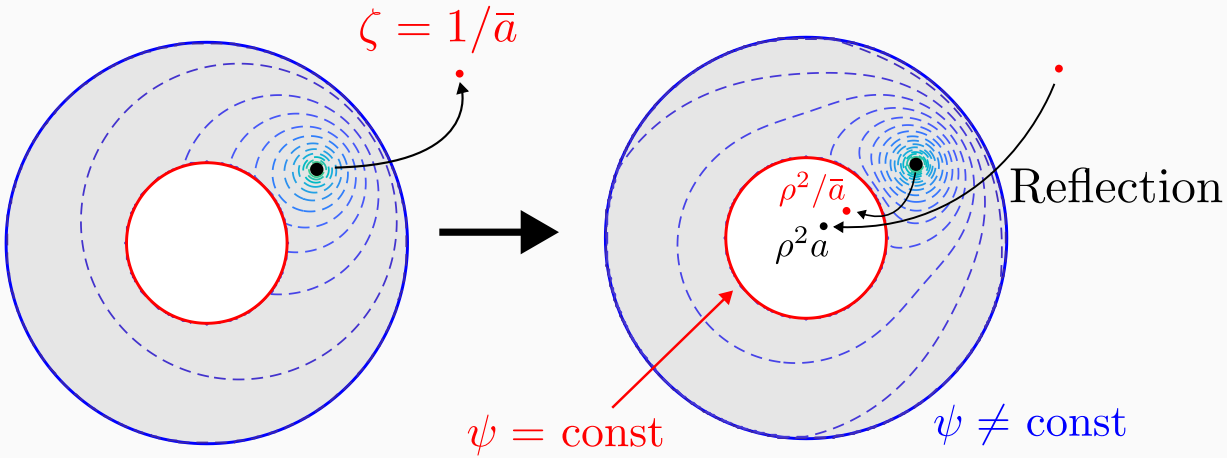


Construction of the complex potential

- Next include the images of $\zeta = a$ and $\zeta = 1/\overline{a}$ in the inner circle.
- Image in the unit circle: $\zeta = 1/\overline{a}$. Image in the inner circle: $\zeta = \rho^2/\overline{a}$.

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}}(\log(\zeta - a) - \log(\zeta - 1/\overline{a}) \\ - \log(\zeta - \rho^2/\overline{a}) + \log(\zeta - \rho^2 a)). \tag{8}$$

- The boundary condition is now satisfied on C_1, but not on C_0.

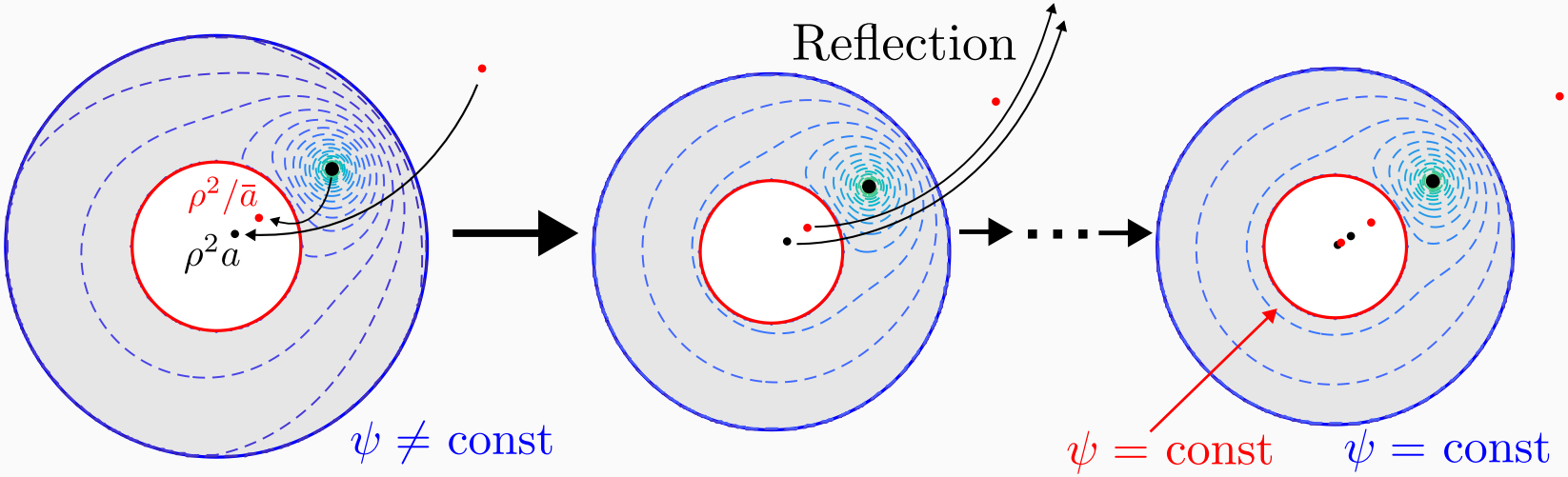


Construction of the complex potential

- Next take images in C_0.

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}}(\log(\zeta - a) - \log(\zeta - 1/\overline{a}) \\ - \log(\zeta - \rho^2/\overline{a}) + \log(\zeta - \rho^2 a)) \\ + \log(\zeta - a/\rho^2) - \log(\zeta - 1/\rho^2\overline{a}) \tag{9}$$

- Repeating this procedure, the two boundary conditions are satisfied more and more accurately.



Construction of the complex potential

- Collecting the $+$ and $-$ terms,

$$
\begin{aligned}
w(\zeta) &= \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{(\zeta-a)(\zeta-\rho^2 a)(\zeta-a/\rho^2)(\zeta-\rho^4 a)}{(\zeta-1/\overline{a})(\zeta-\rho^2/\overline{a})(\zeta-1/\rho^2\overline{a})(\zeta-\rho^4/\overline{a})} \cdots \right) \\
&= \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{(\zeta-a)(\zeta-\rho^2 a)(\zeta-\rho^4 a)}{(\zeta-1/\overline{a})(\zeta-\rho^2/\overline{a})(\zeta-\rho^4/\overline{a})} \cdots \right) \\
&+ \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{(\zeta-a/\rho^2)(\zeta-a/\rho^4)}{(\zeta-1/\rho^2\overline{a})(\zeta-1/\rho^4\overline{a})} \cdots \right) \\
&= \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{(\zeta-a)(1-\rho^2 a/\zeta)(1-\rho^4 a/\zeta)}{(\zeta-1/\overline{a})(1-\rho^2/\overline{a}\zeta)(1-\rho^4/\overline{a}\zeta)} \cdots \right) \\
&+ \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{(1-\rho^2\zeta/a)(1-\rho^4\zeta/a)}{(1-\rho^2\zeta\overline{a})(1-\rho^4\zeta\overline{a})} \cdots \right) + c_0(a,\overline{a})
\end{aligned}
$$

Looking at the numerator and denominator,

$$
\begin{aligned}
&\text{(num.)} \sim -(1-\zeta/a)(1-\rho^2\zeta/a)(1-\rho^2 a/\zeta)(1-\rho^4\zeta/a)(1-\rho^4 a/\zeta)\cdots \\
&\text{(den.)} \sim -(1-\overline{a}\zeta)(1-\rho^2/\zeta\overline{a})(1-\rho^2\zeta\overline{a})(1-\rho^4\zeta\overline{a})(1-\rho^4/\zeta\overline{a})\cdots
\end{aligned}
$$

The Schottky–Klein prime function [1]

In view of the above, we define

$$\omega(\zeta,\alpha) := -\frac{\alpha}{C^2}P(\zeta/\alpha,\rho), \tag{10}$$

$$P(\zeta,\rho) := (1-\zeta)\prod_{n=1}^{\infty}(1-\rho^{2n}\zeta)(1-\rho^{2n}\zeta^{-1}) \tag{11}$$

The complex potential can then be written as

$$w(\zeta) = \frac{\Gamma}{2\pi\mathrm{i}}\log\left(\frac{\omega(\zeta,a)}{|a|\omega(\zeta,1/\overline{a})}\right) = \frac{\Gamma}{2\pi\mathrm{i}}\log\left(\frac{|a|P(\zeta/a,\rho)}{P(\zeta\overline{a},\rho)}\right) \tag{12}$$

- The function $\omega(\zeta,\alpha)$: Schottky–Klein prime function
- It plays a central role in complex analysis and potential theory in multiply connected domains. [1]
- The function $\omega(\zeta,\alpha)$ has the quasi-periodicity familiar from elliptic theta functions.

Properties of $P(\zeta)$

$$P(\zeta,\rho) := (1-\zeta)\prod_{n=1}^{\infty}(1-\rho^{2n}\zeta)(1-\rho^{2n}\zeta^{-1}) \tag{13}$$

Properties

- It has a unique zero in the annulus, at $\zeta = 1$.
- Inversion

$$P(\zeta^{-1},\rho) = -\zeta^{-1}P(\zeta,\rho) \tag{14}$$

- Quasi-periodicity

$$P(\rho^2\zeta,\rho) = -\zeta^{-1}P(\zeta,\rho). \tag{15}$$

A point vortex in the annulus

Complex potential

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{\omega(\zeta, a)}{|a|\omega(\zeta, 1/\overline{a})}\right) = \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{|a|P(\zeta/a, \rho)}{P(\zeta\overline{a}, \rho)}\right) \tag{16}$$

(Why $\mathrm{Im}[w(\zeta)] = \text{const}$ on the boundary)

1. On C_0, $\overline{\zeta} = 1/\zeta$, so

$$\begin{aligned}
\overline{w(\zeta)} &= -\frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{|a|P(\overline{\zeta}/\overline{a}, \rho)}{P(\overline{\zeta}a, \rho)}\right) = -\frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{|a|P(1/\overline{a}\zeta, \rho)}{P(a/\zeta, \rho)}\right) \\
&= -\frac{\Gamma}{2\pi \mathrm{i}} \log\left(|a|\frac{1/\zeta\overline{a} \cdot P(\zeta\overline{a}, \rho)}{a/\zeta \cdot P(\zeta/a, \rho)}\right) = -\frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{P(\zeta\overline{a}, \rho)}{|a|P(\zeta/a, \rho)}\right) \\
&= w(\zeta).
\end{aligned} \tag{17}$$

Hence $\mathrm{Im}[w(\zeta)] = 0$ on $\zeta \in C_0$. The argument on $\zeta \in C_1$ is similar.

Computational method

Computation

- Set the truncation index $N_{\max}$.
- Truncate $P(\zeta, \rho)$ as follows:

$$P(\zeta, \rho) \approx (1 - \zeta) \prod_{n=1}^{N_{\max}} (1 - \rho^{2n}\zeta)(1 - \rho^{2n}\zeta^{-1}) \tag{18}$$

- Define

$$w(z) = \phi + \mathrm{i}\psi = \frac{1}{2\pi\mathrm{i}} \log\left(\frac{\omega(z, a)}{|a|\omega(z, 1/\overline{a})}\right) = \frac{1}{2\pi\mathrm{i}} \log\left(\frac{|a|P(z/a, \rho)}{P(z\overline{a}, \rho)}\right) \tag{19}$$

- The code is in "`03_schottky_klein_annulus.ipynb`"

Streamlines and velocity field

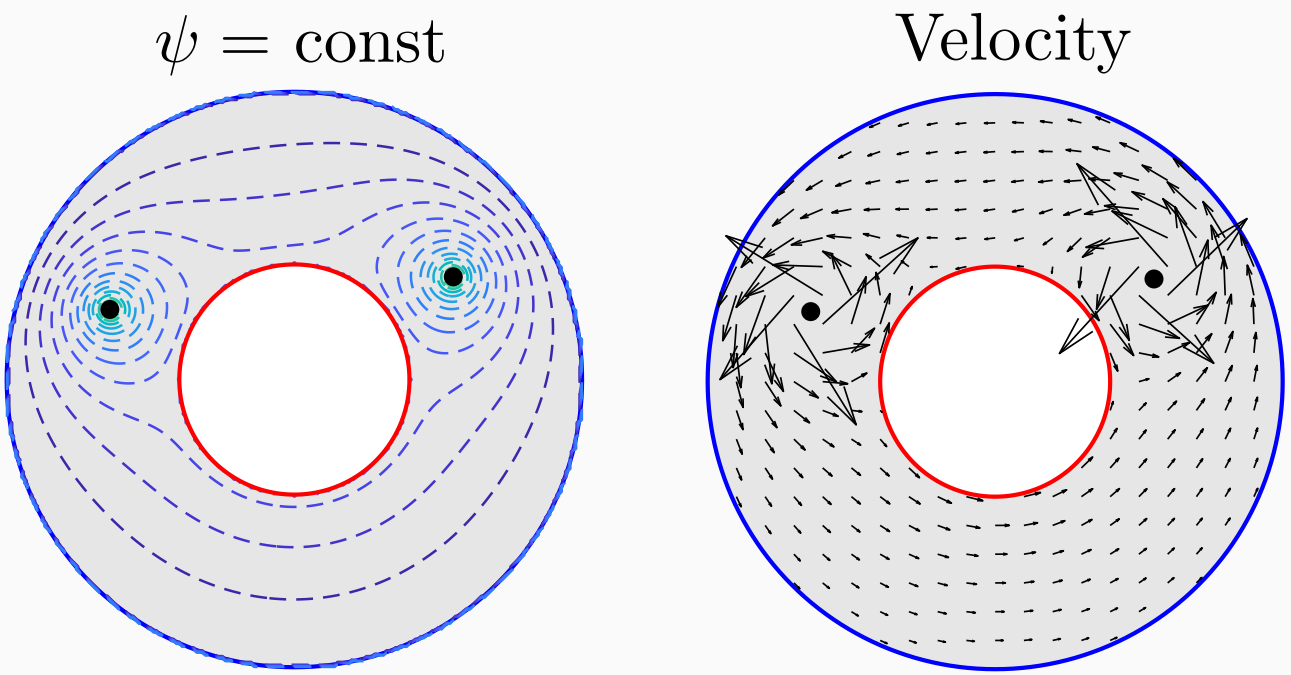


Figure 1: Streamlines and velocity field of two point vortices in an annulus.

Representation for the network and loss

What the network outputs

The unknown is not P. It is the regular part of the vortex,

$$w(\zeta) = \frac{1}{2\pi \mathrm{i}} \log(\zeta - a) + \hat{w}(\zeta), \tag{20}$$

with $a = 0.5$ and $\rho = 0.3$. $\hat{w} = u + \mathrm{i}v$ is an ordinary MLP, $(x, y) \mapsto (u_\theta, v_\theta)$, with $\tanh$ activations. In the notebook: width 64, depth 5.

The logarithm is written by hand. The network is not asked to fit the singularity at $\zeta = a$. Nothing in the architecture forces $u + \mathrm{i}v$ to be holomorphic. The code is `03_vortex_hat_w.ipynb`.

Cauchy–Riemann as a loss

Holomorphicity of $\hat{w}$ is the pair

$$\partial_x u = \partial_y v, \qquad \partial_y u = -\partial_x v. \tag{21}$$

Both derivatives come from `autograd`, as on Day 1. At collocation points in the annulus,

$$\mathcal{L}_{\mathrm{CR}} = \mathrm{mean}\big((\partial_x u - \partial_y v)^2 + (\partial_y u + \partial_x v)^2\big). \tag{22}$$

What this term can do

It penalises a non-holomorphic $\hat{w}$. It does not make the boundary circles into streamlines.

Constant imaginary part on each circle

$$\operatorname{Im} w = -\frac{1}{2\pi} \log |\zeta - a| + v_\theta(\zeta). \tag{23}$$

On $|\zeta| = 1$ and on $|\zeta| = \rho$, the loss is the variance of $\operatorname{Im} w$. The two constants are not required to coincide.

Gauge

The mean of $\operatorname{Im} w$ on the outer circle is set to 0, and the mean of u_θ there is set to 0. The constant on the inner circle is predicted, not prescribed.

The product does not enter this loss.

Circulation on the inner circle

$\hat{w}$ is single-valued, so it contains no term $c \log \zeta$. The loss still records the period

$$\oint_{|\zeta|=\rho} \mathrm{d}u_\theta, \tag{24}$$

which stays at the size of the quadrature.

The explicit logarithm has circulation 0 on $|\zeta| = \rho$ and circulation 1 on $|\zeta| = 1$, because a lies between the two circles. A holomorphic $\hat{w}$ does not change those two numbers.

Check

What is being compared

1. Truncate $P(\zeta, \rho)$ and form w. This is the reference, after training.
2. Build w from $\log(\zeta - a)$ and the trained $\hat{w}$.
3. Compare $\operatorname{Im} w$, and the circulation on each circle.

One run, $\rho = 0.3$, $a = 0.5$

After 3000 steps the relative L^2 error of $\operatorname{Im} w$ is about 0.03. The inner constant is about 0.111, against 0.110 from the product.

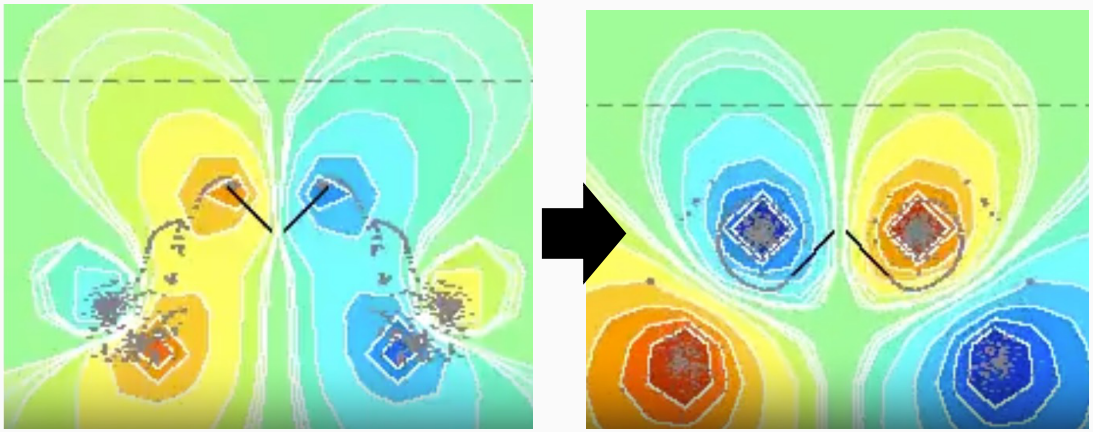
Conclusions

- One hole is an annulus. The reflections are the product $P(\zeta, \rho)$, cut at $N_{\max}$.
- The unit test is the inversion identity at sample points.
- The point vortex is the test case: $\operatorname{Im} w$ is constant on each circle.
- The network learns $\hat{w}$, not P. The product is the check after training.
- Circulation around the inner circle is 0. Around the outer circle it is 1.

Summary

- We introduced a function $P(\zeta, \rho)$ to solve a problem in doubly connected domains.
- We have solved a potential problem, but it can be used for solving many problems in such domains.

Insect hovering



https://fluid.hiroshima-u.ac.jp/%E7%A0%94%E7%A9%B6/

Figure 2: Streamlines and velocity field of two point vortices in an annulus.

References

[1] Crowdy, D., 2020. Solving Problems in Multiply Connected Domains. CBMS-NSF regional conference series in applied math, Society for Industrial and Applied Mathematics. URL: https://books.google.co.jp/books?id=A5hSzQEACAAJ.

References (Japanese books)

- T. Sakajo, *Mathematical Aspects of Vortex Motion*, RCIM
- I. Imai, *Fluid Dynamics* (Vol. 1), Physics Series 14
- I. Imai, *Complex Analysis and Fluid Dynamics*, Nippon Hyoronsha